

Matenchi Kagiso

Brightlearn Exercise 1

SQL Fundamentals: SELECT & Filtering

1. Retrieve all columns from the employees table.

```
SELECT *  
FROM employees;
```

output

id	first-name	last-name	department	Salary	Hire date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-08	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. Find all unique departments.

```
SELECT DISTINCT Department  
FROM employees;
```

output

department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending

Expected columns:

first_name last_name

```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

↓ output

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.

```
SELECT ID_NUM, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

↓ output

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. Find employees in the IT department.

```
SELECT Id, first-name, last-name  
FROM employees  
WHERE department = 'IT';
```

↓ output

Id	first-name	last-name
1	Alice	Green
5	Eva	Naido

6. Find employees in finance with salary > 60000.

```
SELECT Id, first-name, last-name, department,  
       salary  
FROM employees  
WHERE department = 'finance' AND salary > 60000;
```

↓ output

Id	first-name	last-name	department	salary
3	Cathy	Zulu	finance	65000

7. Find employees in HR or Marketing

```
SELECT Id, first-name, last-name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

↓ output.

Id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in IT.

```
SELECT Id, first-name, last-name, department
FROM employees
WHERE NOT department = 'IT';
```

↓ output.

Id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	finance
4	David	Mokoena	Marketing

1. Find employees in IT, HR, or finance using IN.

```
SELECT Id, first-name, last-name, department
FROM employees
WHERE department IN ('HR', 'finance', 'IT');
```

↓ output

id	First_name	Last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg

SELECT id, first_name, last_name, department,
Salary, city

FROM employees

WHERE salary > 65000 AND city = 'Johannesburg'

↓ output

id	first_name	last_name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg